Flex BSC SW User Manual

SW Name	**Flex BSC**
Project	**S-Frame / JF1 Rack System**
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Revision History

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1. Introduction

This document contains guides to operating BSC.

1.1. Glossary

Terms	*Description*
BSC	Battery Section Controller *It normally means the SW that operates battery system.*
BSC IPC	Industrial PC *HW where BSC SW is installed*
Rack	*The minimum communication and charge/discharge units. It consists of BPU, RBMS and Packs*
BPU	Battery Protection Unit
RBMS	Rack Battery Management System *Control device (SW & HW) that is installed in BPU.*
EMS	Energy Management System *SW that operates BESS including BSC and PCS. It is a customer scope.*
PCS	Power Control System

1.2. Communication

To operate BSC, two network configurations are required, internal and external.

DHCP feature for the internal network is necessary for BSC to communicate to RBMSs.

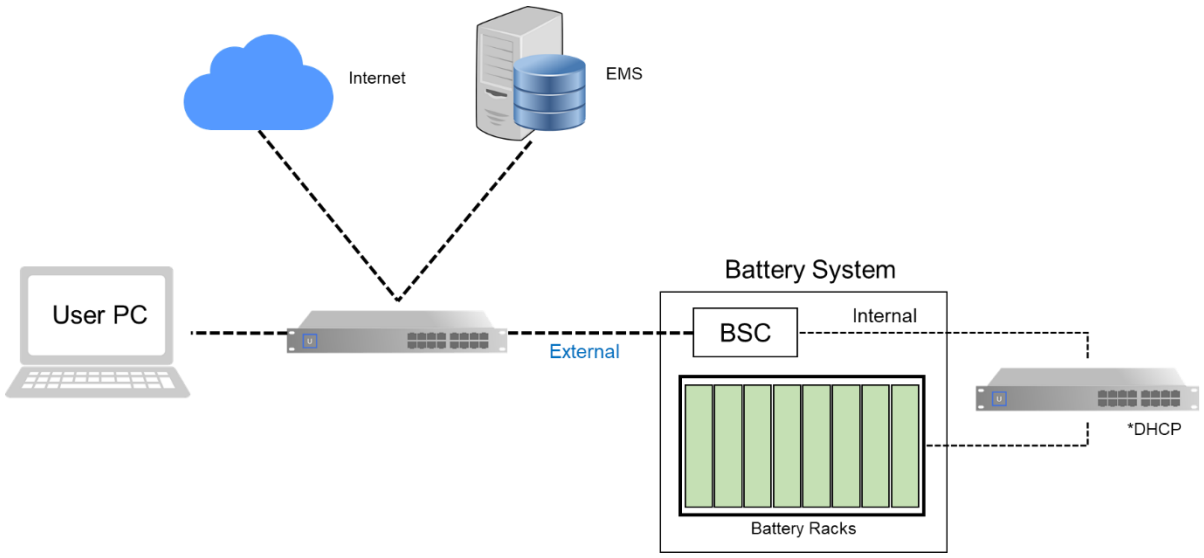


Figure 1. Comm. Configuration

1.3. Setup

1) BSC IP setting

CONFIDENTIAL

In order to communicate with a upper controller such as EMS or PCS or to monitor BSC data through UI programs on a user PC, the external network IP address of BSC must be set.

1-1) Type settings in the search bar to open setting window

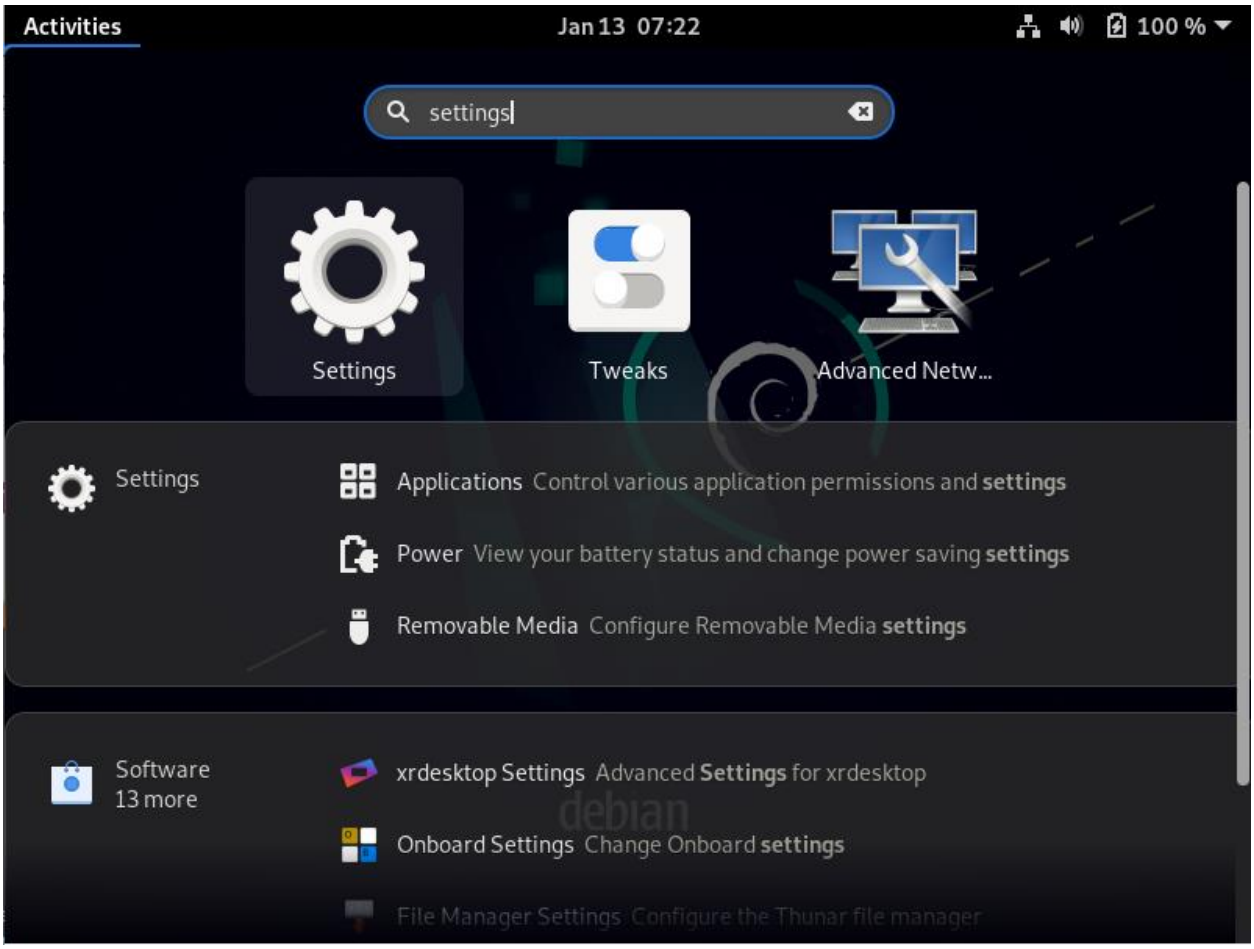


Figure 2. Settings

1-2) Check Wired activation status in Network

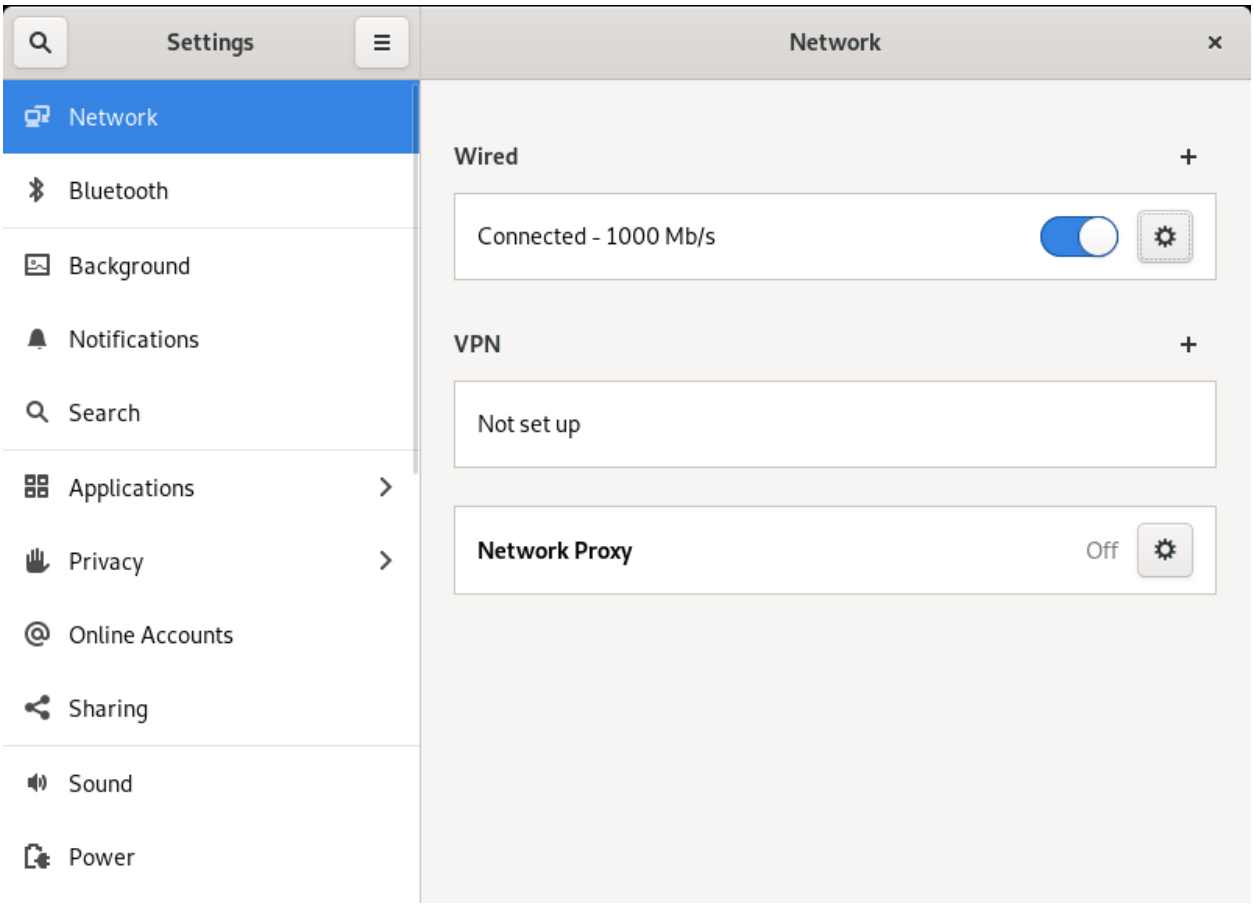


Figure 3. Network

1-3) Select the mac address in the network Identity tab.

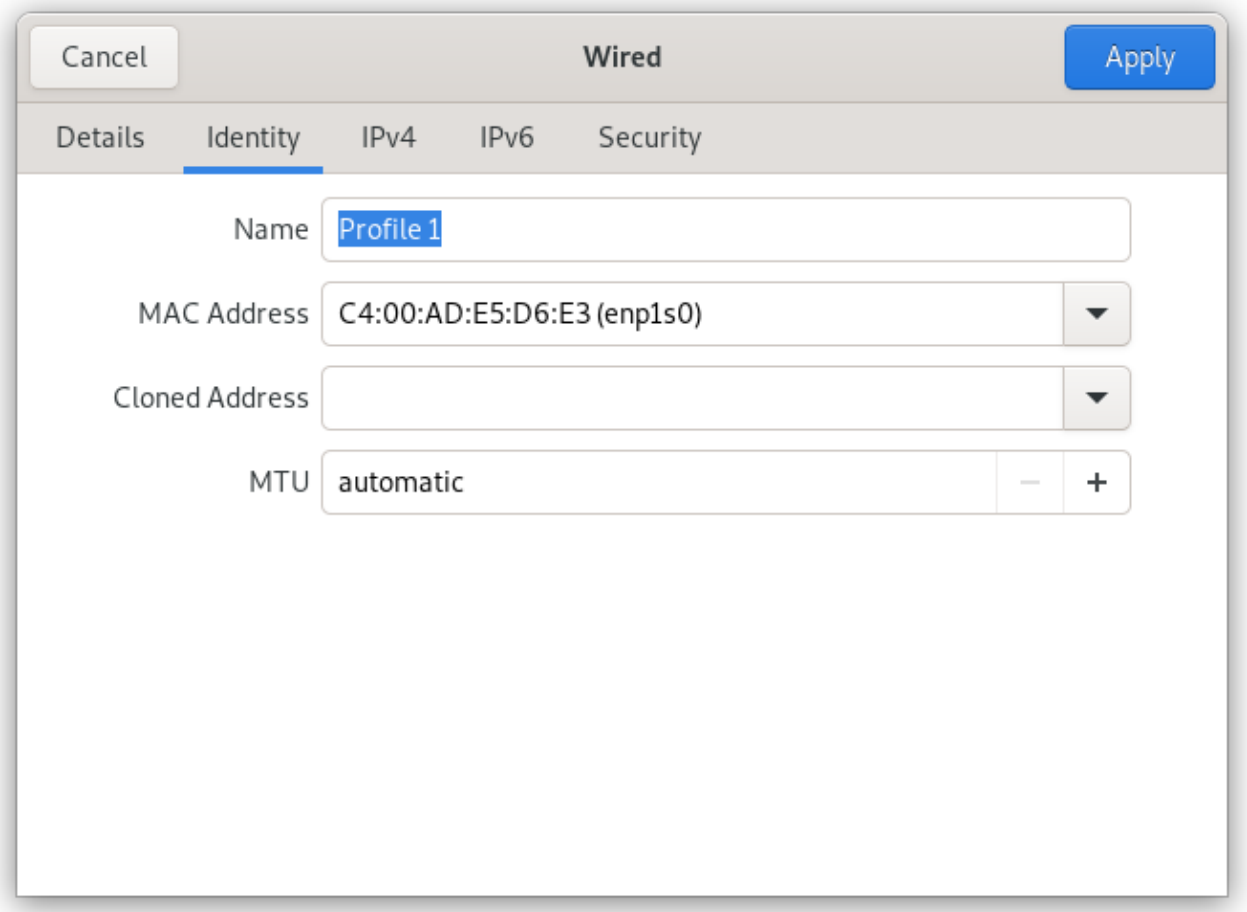


Figure 4. Mac Address

* If there are multiple network cards and the Mac address not set, the same settings may be applied to all network devices.

1-4) Set IPv4 in the detailed settings manually

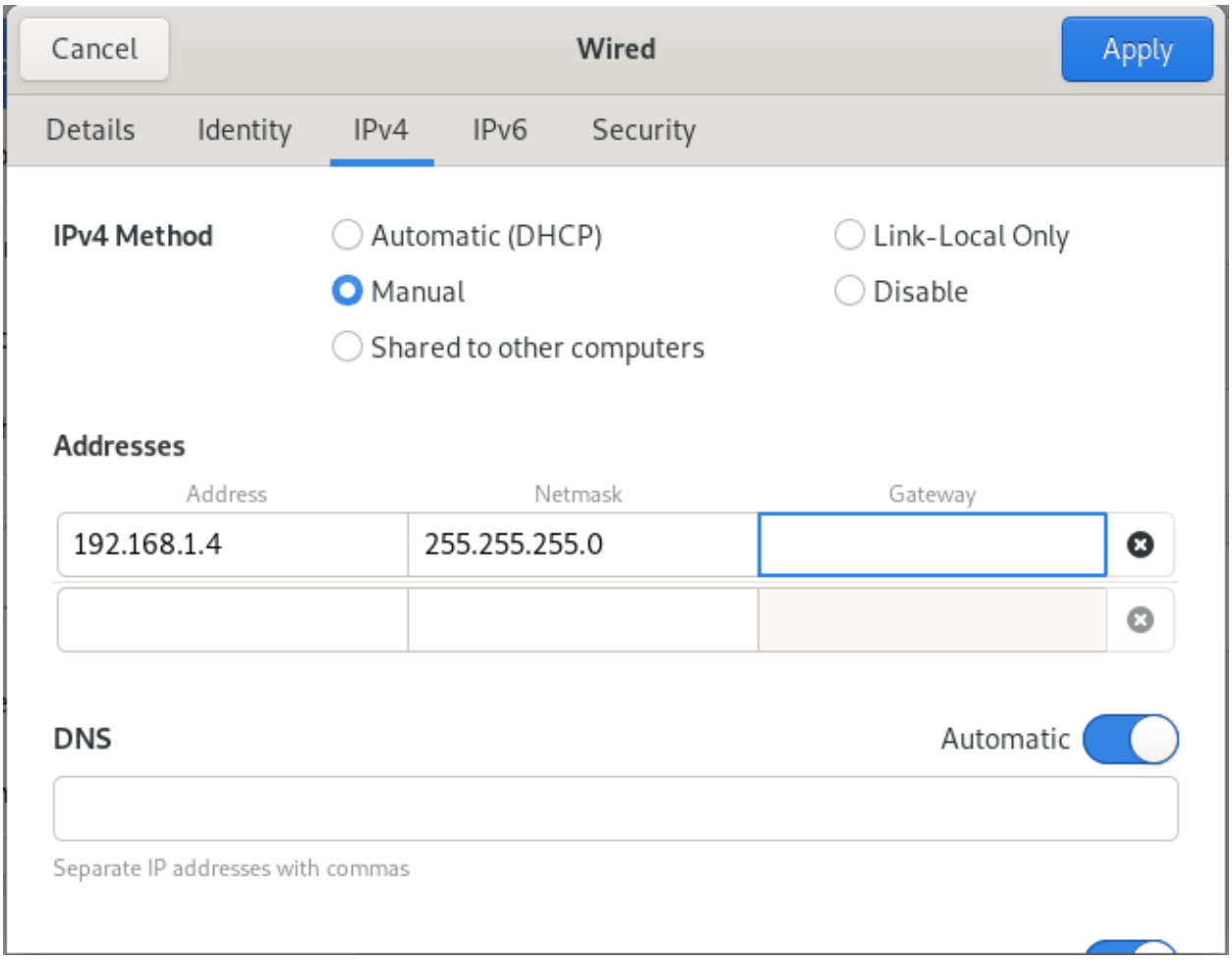


Figure 5. IPv4

2) RBMS IP setting

BSC communicates with RBMS through Modbus TCP and must know the IP address of all RBMS.

Since the RBMS is supposed to receive its own IP address from the network switch in the DHCP method, the switch for the internal network must include the DHCP function.

The Mac address QR code label is attached to the front of the RBMS board, and the network switch must be set to grant an IP to this Mac address.

For more details on network switch setting, please contact the representative of the SI company or for network switch provided by LGES, refer to the LGES network setup guide.

3) User PC

To change or monitor the settings of BSC, a separate user PC must be prepared. The user PC must have a browser such as Chrome/MS Edge installed.

1.4. SW manual installation

To manually install SW, you must transfer the file to be installed to the BSC PC and enter the installation command through the terminal. In general, WinSCP and Putty apps are widely used and recommended.

1.4.1. File transfer

When you run WinSCP, the window below appears.

Enter the BSC PC's connection IP, user account, and password and log in.

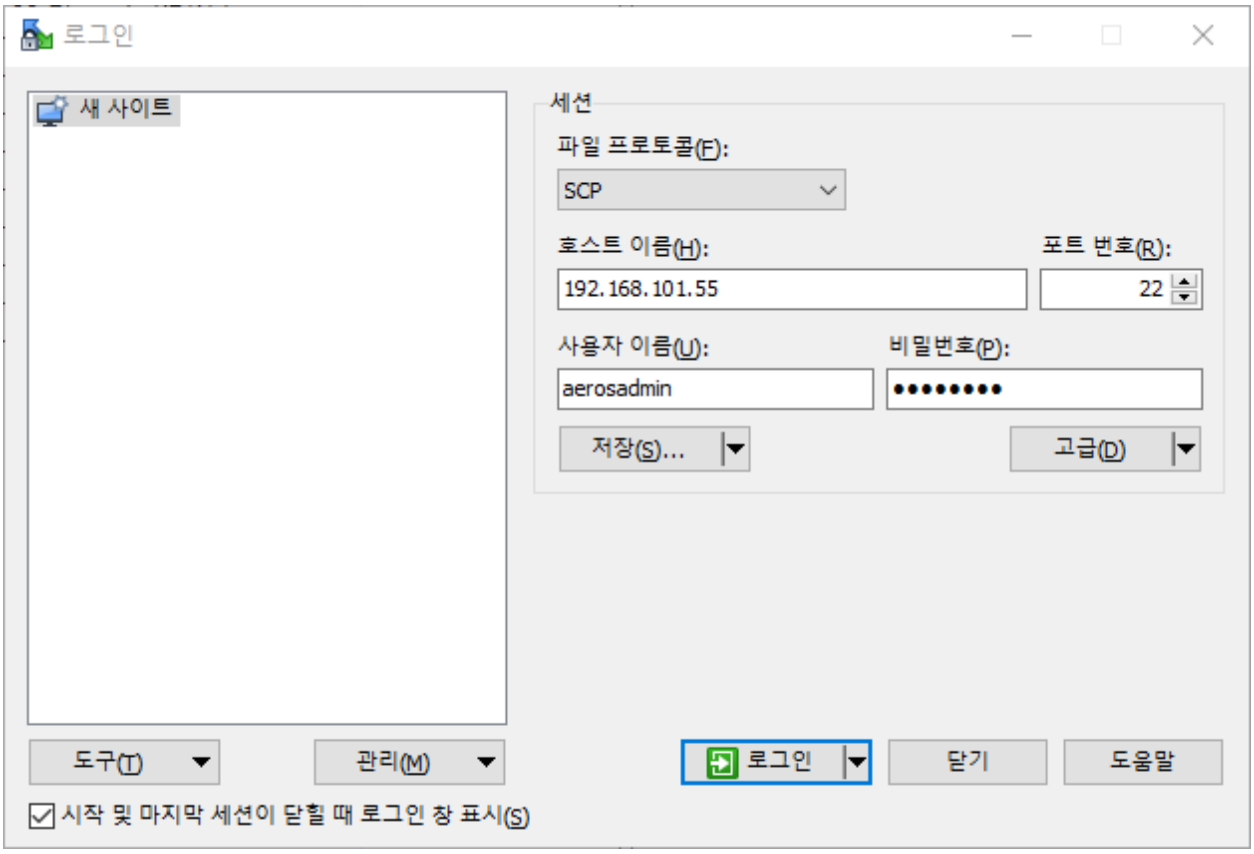


Figure 6. WinSCP Log-in

If you successfully log in, a list of the user PC's folders on the left and the BSC PC's folders on the right will appear as shown below.
You can transfer desired files in both directions using the drag and drop method.

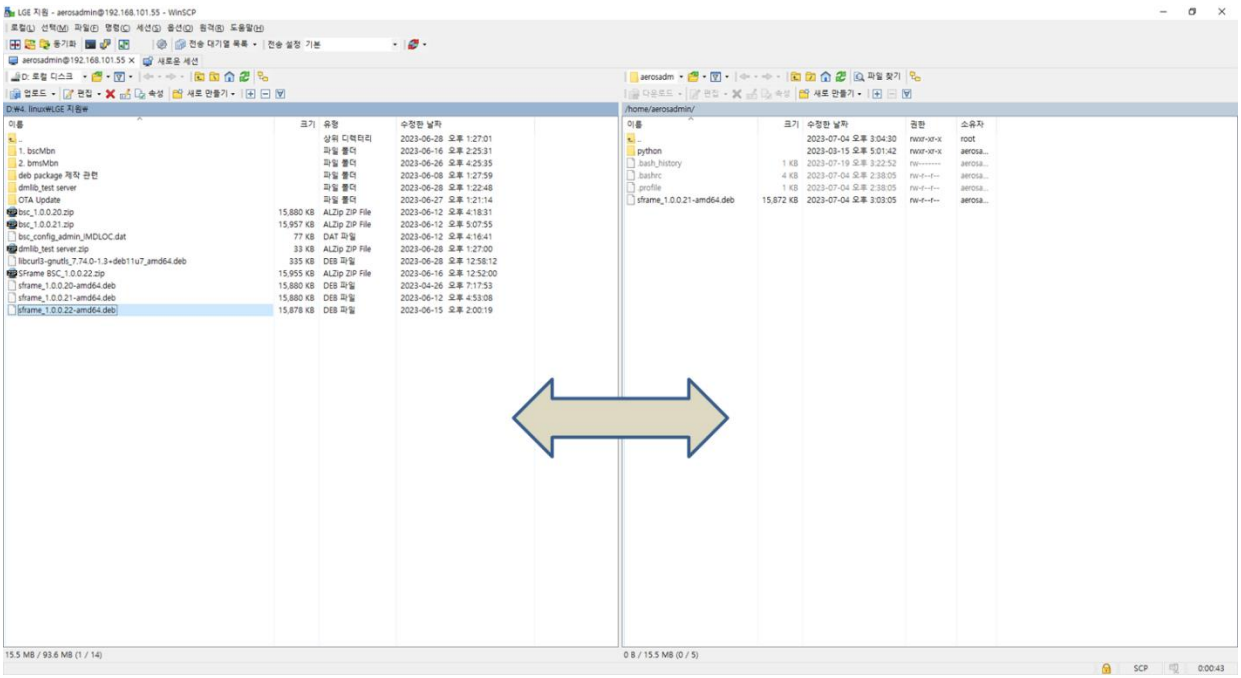


Figure 7. File Transfer

1.4.2. SW Installation

To install SW, you must enter commands through the terminal.

For terminal access, you can run Putty from the WinSCP menu.

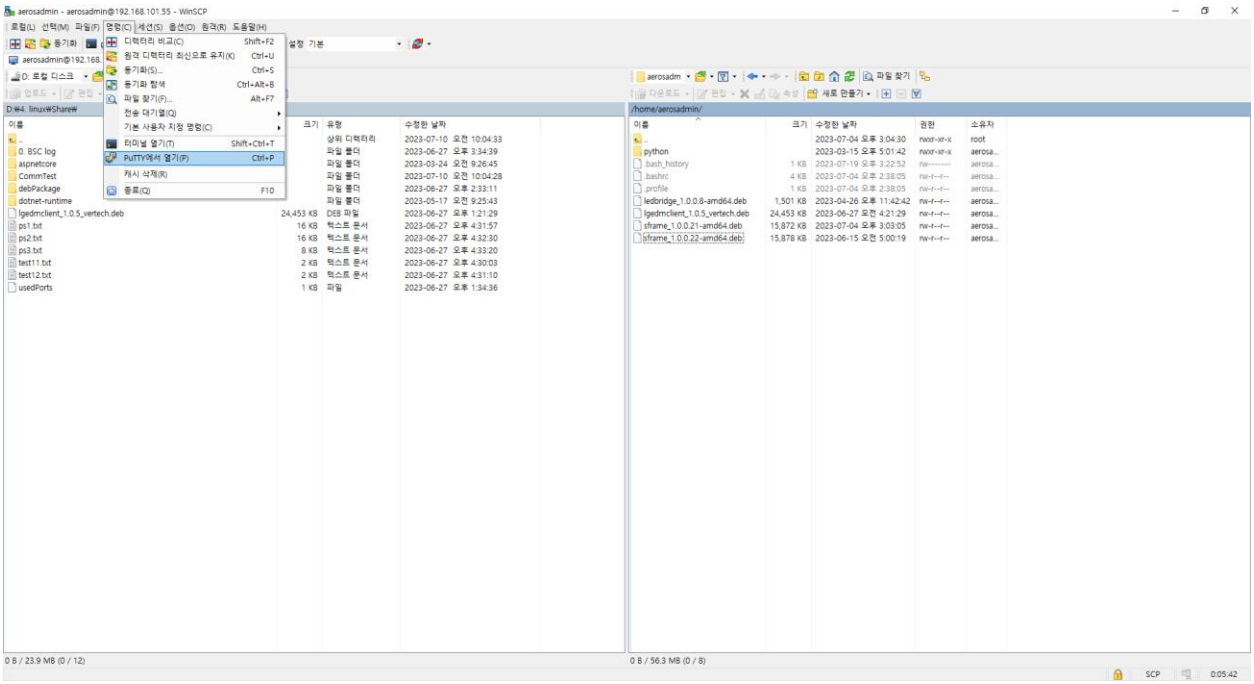


Figure 8. Executing Putty

Input the password to establish terminal connection.

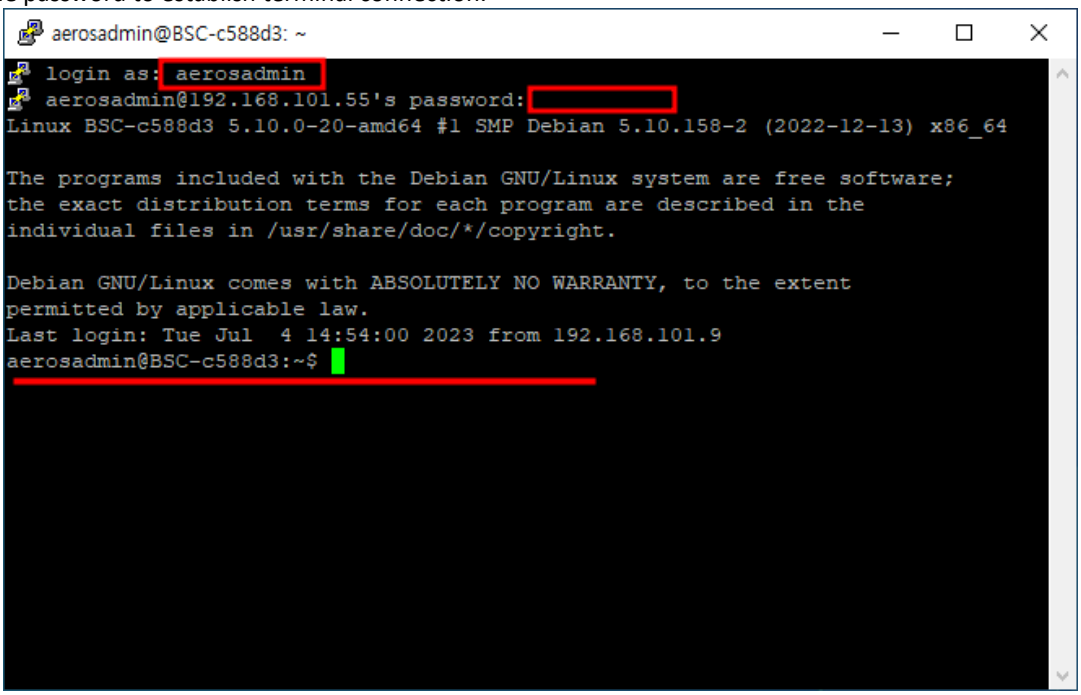


Figure 9. Terminal Connection

1) Installation
   If you type the following, the installation will proceed and complete.
   sudo dpkg -i flexbsc_x.x.x.x-amd64.deb

```
lges@debian:/mnt/shared/debPackage/bsc$ sudo dpkg -i flexbsc_1.0.1.4-amd64.deb
Selecting previously unselected package flexbsc.
(Reading database ... 146907 files and directories currently installed.)
Preparing to unpack flexbsc_1.0.1.4-amd64.deb ...
Unpacking flexbsc (1.0.1.4) ...
Setting up flexbsc (1.0.1.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/flexbsc.service → /e
tc/systemd/system/flexbsc.service.
lges@debian:/mnt/shared/debPackage/bsc$ ^C
lges@debian:/mnt/shared/debPackage/bsc$
```

Figure 10. Installation

2) Check service running
   Type the following to check active status.
   sudo systemctl status flexbsc

```
lges@debian:/mnt/shared/debPackage/bsc$ sudo systemctl status flexbsc
● flexbsc.service - Flex BSC
     Loaded: loaded (/etc/systemd/system/flexbsc.service; enabled; vendor prese>
     Active: active (running) since Sat 2023-09-09 14:36:47 KST; 8s ago
   Main PID: 2150 (Flex_BSC)
      Tasks: 98 (limit: 2321)
     Memory: 431.9M
        CPU: 3.846s
     CGroup: /system.slice/flexbsc.service
             ├─2150 /usr/sbin/lges/gen3bsc/Flex_BSC
             ├─2168 ./Flex_BSC -t Broker
             ├─2190 ./Flex_BSC -t Historian
             ├─2210 ./Flex_BSC -t Gen3BSC.Interface -a S1
             └─2228 ./Flex_BSC -t Gen3BSC.Controller -a S1

Sep 09 14:36:47 debian systemd[1]: Started Flex BSC.
Sep 09 14:36:50 debian Flex_BSC[2168]: Hosting environment: Production
Sep 09 14:36:50 debian Flex_BSC[2168]: Content root path: /usr/sbin/lges/gen3bs>
Sep 09 14:36:50 debian Flex_BSC[2168]: Now listening on: http://[::]:5000
lines 1-18/18 (END)
```

Figure 11. Checking status

3) Etc

If it is not processed as above, you can uninstall or terminate the service with the command below.

3-1) Remove

sudo dpkg -r flexbsc

3-2) Stop service

sudo systemctl stop flexbsc

3-3) Start service

sudo systemctl start flexbsc

2. Starting and Stopping the Battery Section

2.1. Starting BSC

When the PC boots up, BSC runs automatically. If BSC is not running, re-check the *Setup*

The BSC program runs in the background as a server program.

For monitoring, To monitor the battery system, you can access the BSC via a web browser. See 3. UI

Caution) Do not operate other S/W in the BSC PC while the battery system is in operation, as it may degrade the performance of the BSC S/W.

2.2. Starting the battery section

1) Follow the procedure below to operate the battery system.

2) Turn on the power switches of all BPUs.

3) Close the disconnect switches of all BPUs. (Only for BPUs with a disconnect switch.)

4) BSC Turn on the PC and automatically run BSC.

5) When BSC is executed, it automatically tries to connect with BMS. (Transition to *Not Initialized* state in case of failure)

6) If communication with BMS is normally connected, it transitions to *Normal* state.

2.3. Stopping the battery section

1) To stop the battery system for maintenance, etc., follow the procedure below.

2) End charging/discharging.

3) Stop the battery system by disconnecting from the BMS with the command of the upper controller.

4) Check the LEDs to see if all BMS are in their initial state, and turn off the power of the BSC PC to terminate BSC.

5) Open the disconnect switches of all BPUs. (Only for BPUs with a disconnect switch.)

6) Turn off the power switches of all BPUs.

2.4. Connect and disconnect communication

In order to operate the battery system normally, communication between BMS, BSC and Controller must be properly performed.

2.4.1. Connection to Controller

BSC communicates with the controller using Modbus-TCP/IP. At this time, the BSC serves as a Modbus server, and the Modbus port 502 is opened as soon as the BSC SW starts. After the Modbus port is open, the controller can connect with the BSC when it is ready to start. After the controller normally connects to the BSC, the BSC can operate according to the commands of the controller.

2.4.2. Connection to BMS

BSC and BMS communicate using Modbus-TCP.

When the controller sends "Start" command to BSC, BSC changes state to *Initializing* state and starts communication with RBMS.

The BSC accesses the RBMS and assigns an ID so that the RBMS can diagnose and transit to the next state. If normal is confirmed, BSC turns to normal state.

When all RBMS transit to the normal or partially allowed state, the BSC state transits to the *Normal* state.

Note) When the BSC state changes to Normal, the controller must send a Close signal to start charging/discharging.

When the Close signal is received from the controller, considering the voltage deviation, BSC selects racks, and close the contactors of the racks sequentially to make it online.

Note) Some racks cannot close a contactor due to a voltage difference between racks or a fault.

2.4.3. Disconnection from BMS

When the controller sends a "Stop" command to the BSC, the BSC sends a reset command to all RBMS and turns to *Not Initialized*.

Caution) Before disconnecting from the BMS, make sure to stop charging/discharging otherwise the stop command is rejected during charging/discharging.

2.5. **Basic things to be checked**

2.5.1. Before closing contactors

When the value of *BSC Status* is *Normal (Offline),* check the following things.

1) Make sure that there are no warning and fault alarms.

2) Make sure that every BPU's disconnect switch is closed. (Only for the BPU that has the disconnect switch.)
3) Make sure that there is no battery rack that is out of the normal ranges of the key operating conditions, including battery temperature, voltage and SOC (State Of Charge).

2.5.2. During closing contactors

When the value of *BSC Status* is *Normal (Offline)*, check the following things.

1) Wait until all relay closed monitoring the value of *Relay Control* to be *Off*.
   Caution) Every BPU is closing their contactors and charging or discharging the battery section may destroy their contactors.

2.5.3. After closing contactors

When the value of *BSC Status* is *Normal (Online)*, check the following things.

1) The value of *Number of offline racks* should be 0(zero). This indicates the number of problematic battery racks that are unable to charge and discharge.
2) No alarm message should appear in the *Diagnosis view on the home screen.*

2.6. **BSC state**

2.6.1. BSC State Definition

State	Detail explanation
Not Initialized	- Initial state - The state in which communication with the RBMS is cut off
Initializing	- The state initializing the communication between BSC-RBMS
Normal	- Normal state after completing communication initialization between BSC-RBMS. - Charging/discharging is possible depending on whether a diagnosis has occurred (Charging/discharging is not possible in case of a fault)
NPS (Normal Power Saving)	- When the battery is not in use, the PBMS power is turned off to reduce energy consumption.
Manual	- Mode for manually configured rack charge/discharge operation.

	- Perform diagnostics related to battery operating environment and safety.
Emergency	- When an emergency signal is received from the upper system (Controller: PCS or EMS) or an equivalent diagnosis occurs. - All disconnect switches (if present in the BPU) and main contactors open immediately.
PPS (Protective Power Saving)	- All PBMS are turned off to prevent the battery from being over-discharged. *Note) PPS is a state in which the RBMS directly senses the cell voltage and switches itself.* *It is a state that exists only in RBMS..*

2.6.2. BSC state transition

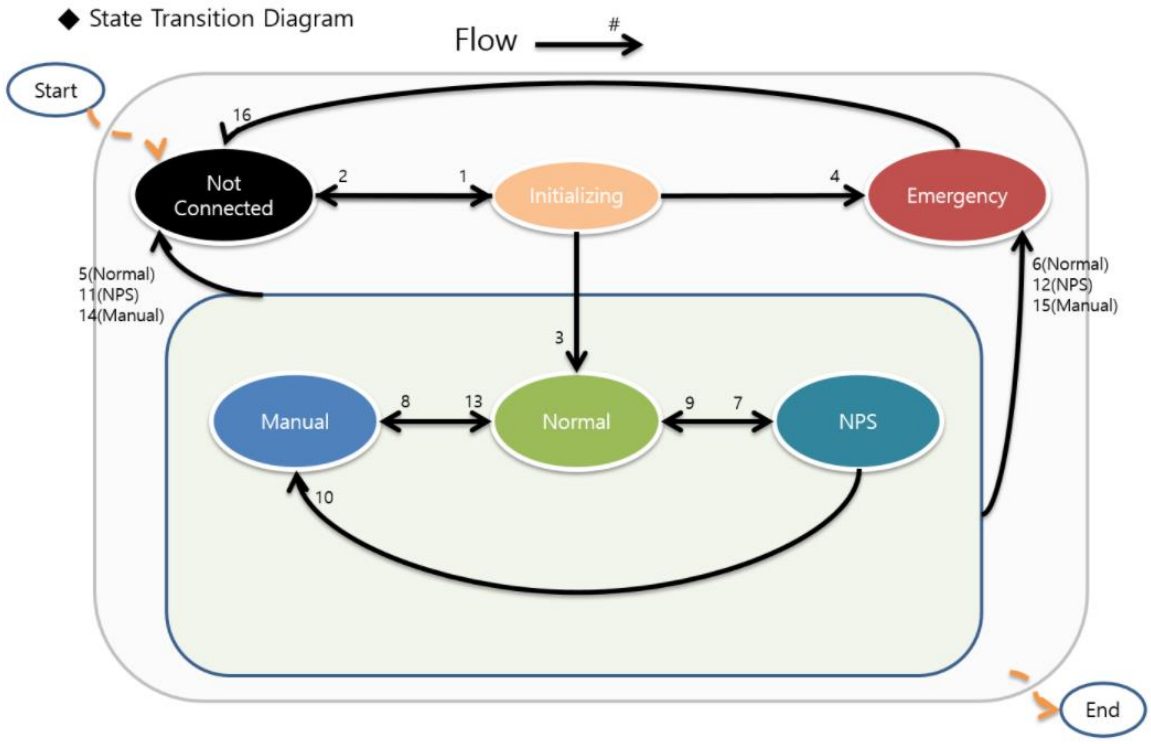


Figure 12 State Transition Diagram

#	From	State transition condition	To
1	Not Connected	Refer to *Starting the battery section*	Initializing

		• In case of receiving 'Start' command from upper controller (Ex. PCS, EMS)	
2	Initializing	• In case of receiving 'Stop' command from upper controller (Ex. PCS, EMS) • If there is a rack that fails to initialize.	Not Connected
3	Initializing	• When all RBMSs have completed initialization	Normal
4	Initializing	• In case of receiving 'Emergency' command from upper controller (Ex. PCS, EMS)	Emergency
5	Normal	• In case of receiving 'Stop' command from upper controller (Ex. PCS, EMS)	Not Connected
6	Normal	• In case of receiving 'Emergency' command from upper controller (Ex. PCS, EMS) • When a diagnosis that transits to emergency mode occurs	Emergency
7	Normal	• When all of the following NPS entry conditions are met and the set time has elapsed a. Charge status is Idle b. When the current of each rack is less than or equal to 3A c. When Max Rack SOC (Online) is lower than NPS entry SOC (set value) : The above conditions assume that at least one online rack exists. d. If there is no rack performing cell balancing	Normal Power Saving
8	Normal	• In case of receiving 'Enter the manual mode' command from upper controller (Ex. PCS, EMS)	Manual
9	Normal Power Saving	• When one or more of the following NPS exit conditions are met a. Charge status is not Idle b. If there are one or more racks with a current of 3A or more. c. In case of receiving 'Relay Control (close / open)' command from upper controller (Ex. PCS, EMS)	Normal
10	Normal Power Saving	• In case of receiving 'Enter the manual mode' command from upper controller (Ex. PCS, EMS)	Manual
11	Normal Power Saving	• In case of receiving 'Stop' command from upper controller (Ex. PCS, EMS)	Not Connected
12	Normal Power Saving	• In case of receiving 'Emergency' command from upper controller (Ex. PCS, EMS) • When a diagnosis that transits to emergency mode occurs	Emergency
13	Manual	• In case of receiving 'Exit the manual mode' command from upper controller (Ex. PCS, EMS)	Normal
14	Manual	• In case of receiving 'Stop' command from upper controller (Ex. PCS, EMS)	Not Connected
15	Manual	• In case of receiving 'Emergency' command from upper controller (Ex. PCS, EMS) • When a diagnosis that transits to emergency mode occurs	Emergency
16	Emergency	• In case of receiving 'Stop' command from upper controller (Ex. PCS, EMS)	Not Connected

3. UI (Web)

3.1. Execution

For BSC monitoring, you can just open the browser and type the address with the specific port to connect to the BSC server. Ex) https://ipaddress:5000.

When Flex BSC is well running, the pain page on the browser appears as shown below

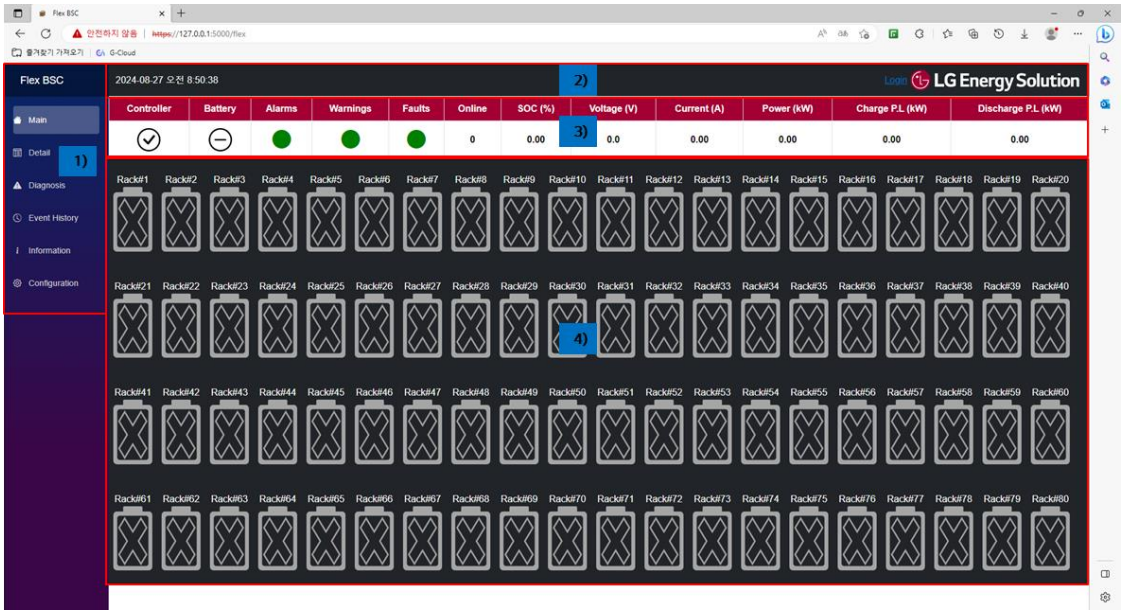


Figure 13. BSC Main Page

1) Side bar: Main, Detail, Diagnosis, Event History, Information, Configuration and etc. are displayed.
2) Header: The time and user authority information are displayed.
3) Battery system summary information: Controller, battery connection status and basic battery system summary information are displayed.
4) Contents: On this screen, you can check the currently reflected system information in real time by selecting the menu bar located on the left.

3.2. Side bar

It is located in the left side of the screen. Depending on your permissions, available functions will appear.

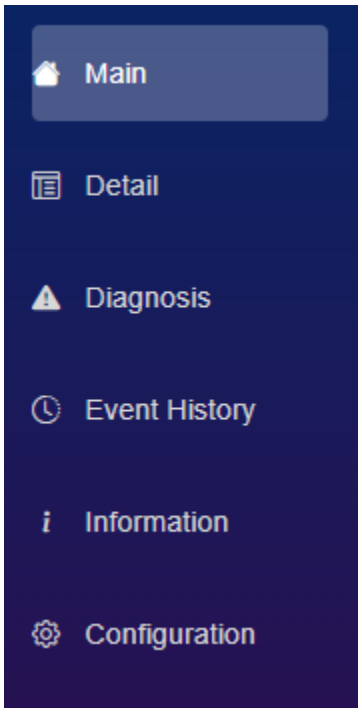


Figure 14. Side bar

3.3. **Header**

It is located on the upper right. The current time and user authority information are displayed.



3.3.1. Permission Change

Through account change, you can get three types of privileges such as User, Admin, and Supervisor for BSC settings and operation. Current permissions can be checked in the upper right corner of the main screen.



Figure 15. Changing Account

If you click the part where the current permission is displayed, a window for entering the password appears as shown below.

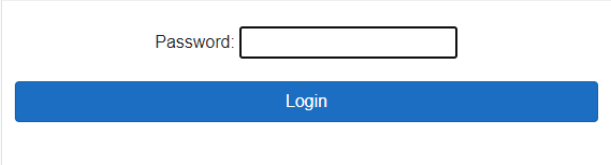


Figure 16. Input Password

If you press the YES button after entering the password for the corresponding authority, the authority automatically changes to the authority corresponding to the password. Through permission change, you can get three permissions such as User, Admin, and Supervisor.

1) *User* : Permission to set and manipulate BSC basic settings such as BMS communication connection, disconnection, and user option setting with initial default user rights
2) *Admin* : Authorization including user privileges and import or export of Admin Option information files
3) *Supervisor* : Authority that includes Admin authority and allows individual setting of each item of Admin Option

3.4. **Battery System Summary Information**

In the upper left corner of the home screen, there is a part where you can check the overall state of the current system.

Controller	Battery	Alarms	Warnings	Faults	Online	SOC (%)	Voltage (V)	Current (A)	Power (kW)	Charge P.L (kW)	Discharge P.L (kW)
✓	✓	●	●	●	24	84.10	752.1	0.00	0.00	1743.38	1743.41

Figure 17. System Status Window

1) Controller, Battery : It represents the communication status of the controller and battery rack connected to the BSC. The image changes as follows depending on the communication status.

 : Connected to communication

 : Loss of communication

2) Alarm, Warning, Fault : Depending on the diagnostic, the image will change as follows:

 : Alarm, Warning, Fault has occurred.

 : Alarm, Warning, Fault has not occurred.

3) Online : The number of racks in all online states
4) SOC(%) : Total SOC of the current battery system (online basis).
5) Voltage(V) : The average voltage of all the online battery racks.
6) Current(A) : The total amount of current in the battery section.
7) Power(kW): The total amount of power in the battery section.
8) Charge P.L(kW): Chargeable Power Limit
9) Discharge P.L(kW): Dischargeable Power Limit

3.5. **Battery System Main View**

On this screen, you can check the currently reflected system information in real time by selecting the menu bar located on the left. - It consists of the Main, Detail, Diagnosis, Event History, Information, and Configuration menus.

3.5.1. Main

You can check the current battery status through each rack image. If you hover your mouse over the rack image, you can check information such as Comm. Status, SOC, and Online status.



Figure 18. Main page (battery rack status)



: A rack that is not communicating with the BSC.



: A rack that is communicating with the BSC, but the DC line of the rack is offline.



: A rack that is communicating with the BSC, and the DC line of the rack is online.



: Warning diagnosis has occurred.



: Fault diagnosis has occurred.

SOC(%): 84.10
SOH(%): 84.10
Current(A): 0.000
Cell Sum. Voltage(V): 752.0589
Battery Charging Status(Enum): Idle
RBMS-BSC Comm LOC: False
Warning Flag: False
Fault Flag: False
DC Line Status(Enum): Open

Figure 19. Rack Image Tooltip Information

항목	상태 / 세부 설명
SOC(%)	Rack SOC
SOH(%)	Rack SOH
Current(A)	The total amount of current in the battery rack..
Cell Sum. Voltage (V)	The sum of all the cell voltages in the battery rack.
Battery Charging/Discharging Status(Enum)	Charging/Discharging Status of the rack (Charging / Discharging / Idle) - Charging: It is currently charging. - Discharging: It is currently discharging. - Idle: It is neither charging nor discharging.
BSC-Rack LOC	BSC-Rack Loss of Communication(True : not connected, False : connected)
Warning Flag	Warning Diagnosis Flag (True : Warning occurred, False : Warning not occurred)
Fault Flag	Fault Diagnosis Flag (True : Fault occurred, False : Fault not occurred)
DC Line Status(Enum)	The connection status of main contactors (Open / Closed)

3.5.2. Detail

Detailed state values of the BSC and detailed information of the battery rack connected to the BSC may be checked. As shown below, it consists of four parts that display BSC information, Battery Rack information, Pack information and Meta Information of racks.

* Depending on the RBMS type, certain data may be added or absent.

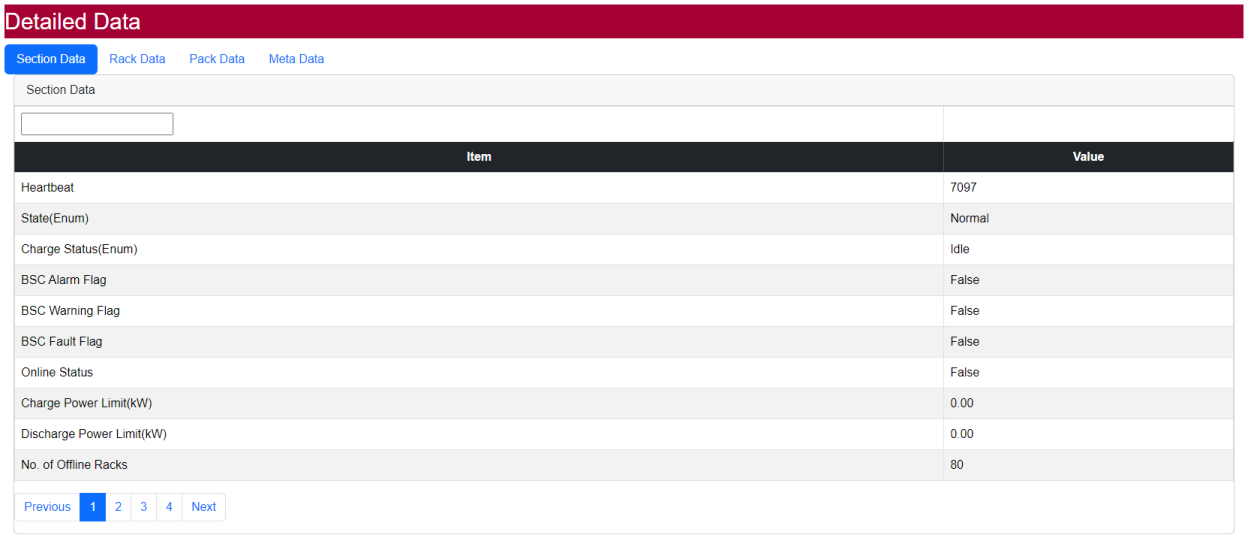


Figure 20. Detail Tab

1) Section Data : Current flow state, Battery diagnosis state, Battery SOC, Power limit, etc. can be checked.

Item	State / Description
Heartbeat	The number that increases by 1 every second that BSC sends.
State(Enum)	See BSC State Definition.
Charge Status(Enum)	Charging/Discharging Status of the BSC (Charging / Discharging / Idle) -. Charging: It is currently charging. -. Discharging: It is currently discharging. -. Idle: It is neither charging nor discharging.
BSC Alarm Flag	BSC Alarm status (True: Alarm occurred, False: Alarm not occurred)
BSC Warning Flag	BSC Warning status (True: Warning occurred, False: Warning not occurred)
BSC Fault Flag	BSC Fault status (True: Fault occurred, False: Fault not occurred)
Online Status	There is an online rack, whether charging/discharging is possible
Charge Power Limit(kW)	Available charge power value of the battery system
Discharge Power Limit(kW)	Available discharge power value of the battery system
No. of Offline Racks	The number of offline racks
No. of Online Racks	The number of online racks
DC Current(A)	The total amount of current of all the online racks
DC Power(kW)	The total amount of DC power of all the online racks
Online Avg. Cell Sum. Voltage(V)	The average voltage of all the online racks
Online System SOC(%)	The system SOC(State Of Charge) of all the online racks
Online Avg. SOH(%)	The average SOH(State Of Health) of all the online racks
Online Avg. Cell Voltage(V)	The average cell voltage of all the online racks
Online Max. Cell Voltage(V)	The maximum cell voltage of all the online racks
Online Min. Cell Voltage(V)	The minimum cell voltage of all the online racks
Online Max. Cell Voltage Rack ID	Rack ID of maximum cell voltage of all the online racks

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Online Max. Cell Voltage Pack ID	Pack ID of maximum cell voltage of all the online racks
Online Max. Cell Voltage Cell ID	Cell ID of maximum cell voltage of all the online racks
Online Min. Cell Voltage Rack ID	Rack ID of minimum cell voltage of all the online racks
Online Min. Cell Voltage Pack ID	Pack ID of minimum cell voltage of all the online racks
Online Min. Cell Voltage Cell ID	Cell ID of minimum cell voltage of all the online racks
Number of recalibration racks	The number of online rack that require recalibration.
TPC Status	Temperature-based Power Control status
SPC Step Charge	SOC-based Power Control charge step (Normal, Step1, Step2)
SPC Step Discharge	SOC-based Power Control discharge step (Normal, Step1, Step2)
VPC Step Charge	Voltage-based Power Control charge step (derating count)
VPC Step Discharge	Voltage-based Power Control discharge step (derating count)
VPC Upper Rack Power(W)	Rack Power when the upper VPC is running
VPC Lower Rack Power(W)	Rack Power when the lower VPC is running
Under Relay Control	Indicates whether the Main contactors are currently being controlled.

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2) Rack Data : Rack level information can be checked.

Item	State / Description
State(Enum)	Depending on the BSC, the RBMS state is changed. See BSC State Definition.
Heartbeat	The number that increases by 1 every second that a Rack sends.
Warning Flag	Warning Diagnosis Flag (True : Warning occurred, False : Warning not occurred)
Fault Flag	Fault Diagnosis Flag (True : Fault occurred, False : Fault not occurred)
Battery Charging Status(Enum)	Charging/Discharging Status of the rack (Charging / Discharging / Idle) - Charging: It is currently charging. - Discharging: It is currently discharging. - Idle: It is neither charging nor discharging.
DC Line Status(Enum)	Online status - Open: Offline - Closed: Online
SOC(%)	Rack SOC
SOH(%)	Rack SOH
Charge Power Limit(kW)	The maximum current allowed to charge the battery rack.
Discharge Power Limit(kW)	The maximum current allowed to discharge the battery rack.
Cell Sum. Voltage(V)	The total amount of cell voltage
Current(A)	Rack current
Precharge Contactor Status	Precharge Contactor status
Main Contactor Plus Status	Main Contactor Plus status
Main Contactor Minus Status	Main Contactor Minus status
Circuit Breaker Status	Circuit Breaker status
Pack Fan1 Running Status	Pack Fan1 running status
Pack Fan2 Running Status	Pack Fan2 running status
BPU Fan Running status	BPU Fan running status
Fuse Status Feedback	Fuse running feedback
Max. Cell Voltage(V)	Maximum cell voltage in a rack

Min. Cell Voltage(V)	Minimum cell voltage in a rack
Avg. Cell Voltage(V)	Average cell voltage in a rack
Max. Cell Voltage Pack ID	Pack ID of maximum cell voltage in a rack
Max. Cell Voltage Cell ID	Cell ID of maximum cell voltage in a rack
Min. Cell Voltage Pack ID	Pack ID of minimum cell voltage in a rack
Min. Cell Voltage Cell ID	Cell ID of minimum cell voltage in a rack
Max. Pack Temperature(℃)	Maximum pack temperature in a rack
Min. Pack Temperature(℃)	Minimum pack temperature in a rack
Avg. Pack Temperature(℃)	Average pack temperature in a rack
Cell Balancing Status	Cell balancing status (True: Cell balancing, False: Not balancing)
CER Code(Enum)	CER(Critical Event Record) occurred status - Normal: Not occurred - PDVF: Occurred by PDVF - RDVF: Occurred by PDVF
Live Unit Count	Live unit count
Remaining Balancing Time	Remaining balancing time
Count of relay open over specific current	Count of relay open over specific current
SOX recommend Full Charge for SOC recalibration	SOX recommend Full Charge for SOC recalibration - ex) 0x00: No request , 0x01 : SOX recommend Full Charge for SOC Recalibration
Recalibration Count from SOX	The value is incremented by 1 each time RBMS recalibrates the SOC.

3) Pack Data : You can check the Pack Level information of the selected Rack on the Rack Status screen.

Item	State / Description
SOC(%)	Pack SOC
Max Cell SOC(%)	Minimum cell SOC in a pack
Min Cell SOC(%)	Maximum cell SOC in a pack
SOH(%)	Pack SOH
Avg. Cell Voltage(V)	Average Cell Voltage in a pack
Max. Cell Voltage(V)	Maximum cell voltage in a pack
Min. Cell Voltage(V)	Minimum cell voltage in a pack
Max. Cell Voltage Cell ID	the ID of the cell that measured maximum cell voltage in a pack
Min. Cell Voltage Cell ID	the ID of the cell that measured minimum cell voltage in a pack
Avg. Temperature(℃)	Average temperature in a pack
Max. Temperature(℃)	Maximum temperature in a pack
Min. Temperature(℃)	Minimum temperature in a pack
Max Pack Temperature Sensor Chn.	The number of the sensor that measured maximum temperature
Min Pack Temperature Sensor Chn.	The number of the sensor that measured minimum temperature
BMIC# Error Code#1(Hex)	BMIC# Error Code#1
BMIC# Error Code#2(Hex)	BMIC# Error Code#2
Pack Temperature # (℃)	Temperature of sensor# in a pack
PCB Temperature# (℃)	PCB temperature of sensor# in a pack
BMIC# Cell# Cell Balancing Status	Cell balancing status of each cell (True: Cell balancing, False: Not balancing)
BMIC# Cell# Cell Invalid Flag	Cell invalid flag of each cell (True: PBMS malfunction, False: Normal)
Cell # Voltage(V)	Cell voltage in a pack

4) Meta Data : You can check the Meta Data of Racks connected to BSC for each Rack.

Item	State / Description
Rack Mode(Enum)	RBMS operating mode (Normal: normal operating mode, Factory: EOL mode, Service: NV setting mode)
Checksum	Checksum
SW Version	RBMS SW version
Updater Version	Updater version
Boot Selector Version	Boot Selector version
SW Name	RBMS SW name
Serial Number	RBMS Serial Number
HW Version	RBMS HW version
Current Sensor Type(Enum)	Current sensor type - Hall - Network - Hall + Network
CB Trip Current Specification(A)	Current specification when CB trip
Relay Close Current Specification(A)	Current specification when Contactor close
Pack Count	The number of packs in a rack
BMIC Count per Pack	The number of BMIC in a pack
Temperature Sensor Count per Pack	The number of temperature sensors in a pack
Ambient Temperature Sensor Count per Pack	The number of ambient temperature Sensors in a pack
PCB Temperature Sensor Count per Pack	The number of PCB temperature sensors in a pack
Pack Voltage Sensor Count per Pack	The number of pack voltage sensors in a pack
Voltage Sensor Count per Pack	The number of voltage sensors in a pack
Represented PBMS SW Version	Represented PBMS SW version in a rack
Pack Type	Pack type
P#.PBMS HW Version	PBMS HW version
P#.Serial Number	Pack Serial Number

3.5.3. DIAGNOSIS

It displays the current diagnosis in real time.

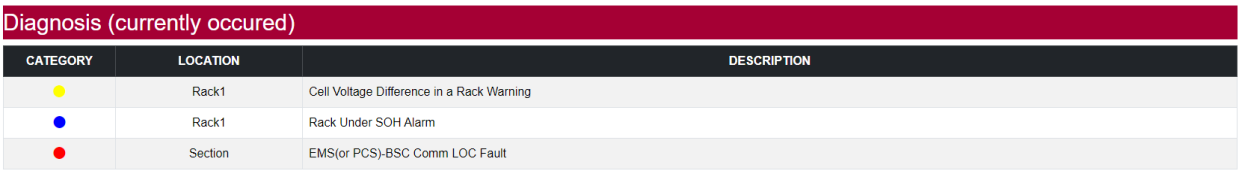


Figure 21. Diagnosis Window

Item	State / Description
CATEGORY	The lamps are displayed, as shown below depending on the diagnostic level. - Alarm : Blue information lamp - Warning : Yellow lamp - Fault : Red lamp
LOCATION	It shows the location where the diagnosis occurred - Section : BSC Level Diagnosis - Rack#No : Rack Level Diagnosis
DESCRIPTION	It shows diagnostic description. Refer to diagnostic document.

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3.5.4. Event History

All events occurring during BSC operation and communication are displayed in the Event History below.

Event History

TIME	CATEGORY	DESCRIPTION
2024-08-27 오후 2:05:10	Diagnosis	Section : Over Offline Rack Warning
2024-08-27 오후 2:05:08	Diagnosis	Rack1 : BSC-RBMS Comm LOC Fault1
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : SBC Fail Safe Redundant Error Fault1
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : SBC RSTB Error Fault1
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Vcca 5V Error Fault1
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Current Sensor Error Fault1
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Frequent Balancing Alarm
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Cycle Warranty Expiration Alarm
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Cell Voltage Difference in a Rack Warning
2024-08-27 오후 2:04:58	Diagnosis	Rack1 : Cell Voltage Difference in a Pack Fault1
2024-08-27 오전 10:55:02	Release	Section : Over Offline Rack Warning released
2024-08-27 오전 10:54:55	Diagnosis	Section : Over Offline Rack Warning
2024-08-27 오전 10:54:51	Command	Close Contactors

Figure 22. Event History

Item	State / Description
TIME	The time the event occurred
CATEGORY	The type of event
DESCRIPTION	Detailed description

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3.5.5. Info

You can check the system information of the PC where BSC is installed, the BSC module version, and license-related information.

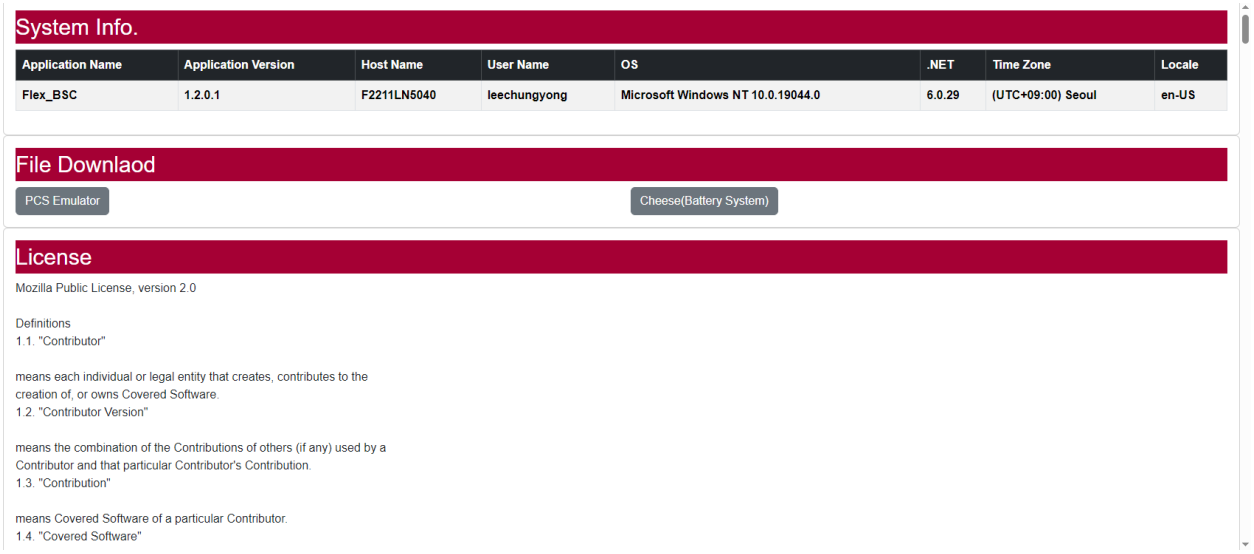


Figure 23. Info Tab

1) SYSTEM INFO

Item	State / Description
Application Name	BSC type (SW Name)
Application Version	BSC Version
Host Name	The name of the PC
User Name	The user name logged-in
OS	OS version
.NET	.NET version
Time Zone	Time zone
Local	Culture information

2) File Download

- PCS Emulator: The PCS emulator can control the BSC when PCS or EMS is not present. **(For LG ES employees only: PW required)**

- Cheese: Cheese is a SW that acts as RBMS when there is no real rack. This SW is used when emulating BSC.

3) License: It contains the module version of the BSC program and various license information used.

3.5.6. Configurator

It is a program that can change the BSC option, and you can check and change the setting values required for BSC operation.

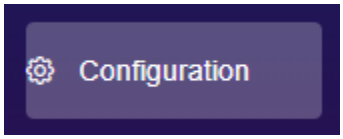


Figure 24. Configurator Icon

1) Common



Figure 25. Configurator - Common

1-1) User Import/Export: You can save or load the value set in User Option to the local PC.

If the password of Supervisor authority is entered, the 'Apply', 'Import' and 'Export' button will appear and additional setting tabs are created, and you can set the appropriate value for the tab.

User Read/Apply : You can read or apply the set values from/to server.

If you click the 'Apply' button, the changed data of the current screen is transmitted to the BSC server and the value is reflected in the server. When the values are reflected in the server, BSC automatically restarts with the changed settings.

Items that can be set vary depending on each account.

2) User Option

Items that can be changed with User permission are displayed.

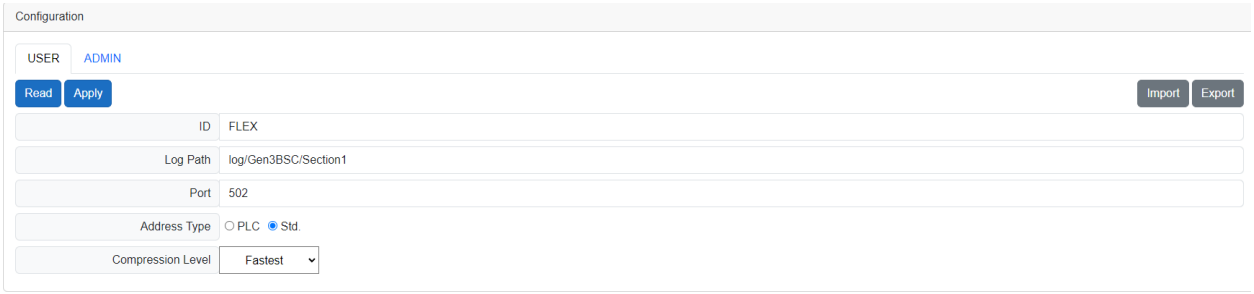


Figure 26. User Option

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2-1) ID: Unique information can be entered to distinguish the site with the BSC unique identifier.

2-2) Logging File Path: This is the log file path while BSC is running.

2-3) Upper Layer Control: You can set the IP, Port, and Address Type of the upper controller (Main Controller – PCS or EMS). Address Type can select Std. and PLC.

- Std.: The address on the map, the address that BSC actually reads/writes are the same. (default)
- PLC: The address on the map, it is 1 greater than the address that BSC actually reads/writes.

2-4) Compression Level: Sets the compression options to be applied when compressing old log files.

3) Admin authority

If you have administrator rights or higher, you will see other configuration tabs. Administrator rights have access to the functions for changing configuration (import/export/read/apply). However, individual settings cannot be changed with administrator rights.

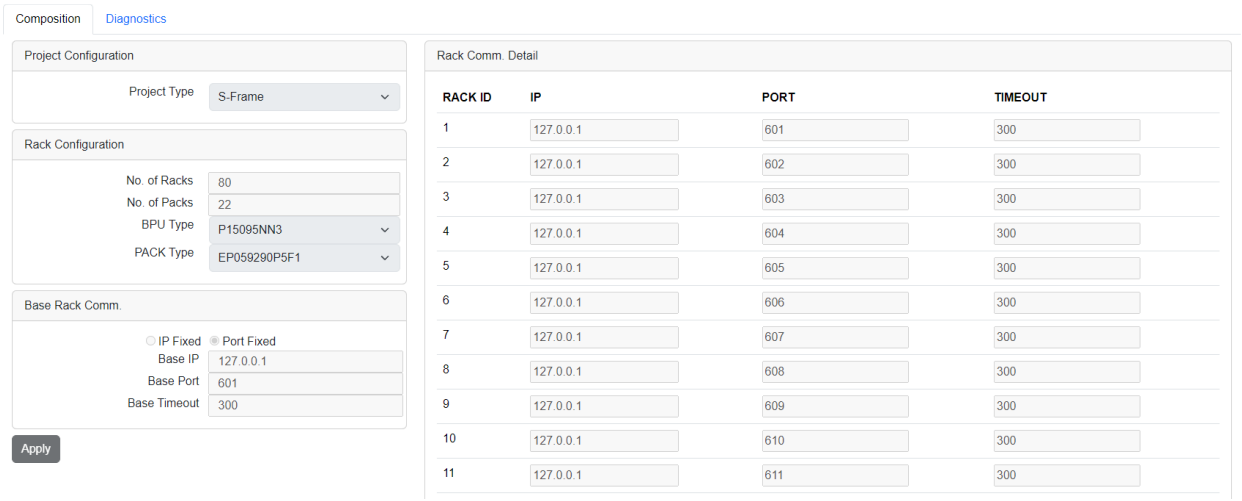


Figure 27. Admin Option

3.5.7. NV Setting (Admin authority)

NV Setting is a function that enables BMS to perform various commands operating in Site Mode. When NV Setting is executed, communication with the existing BSC upper controller is blocked, and when NV work is completed and NV Setting is terminated, upper communication is resumed and BSC is automatically executed.

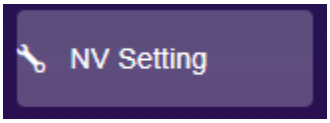
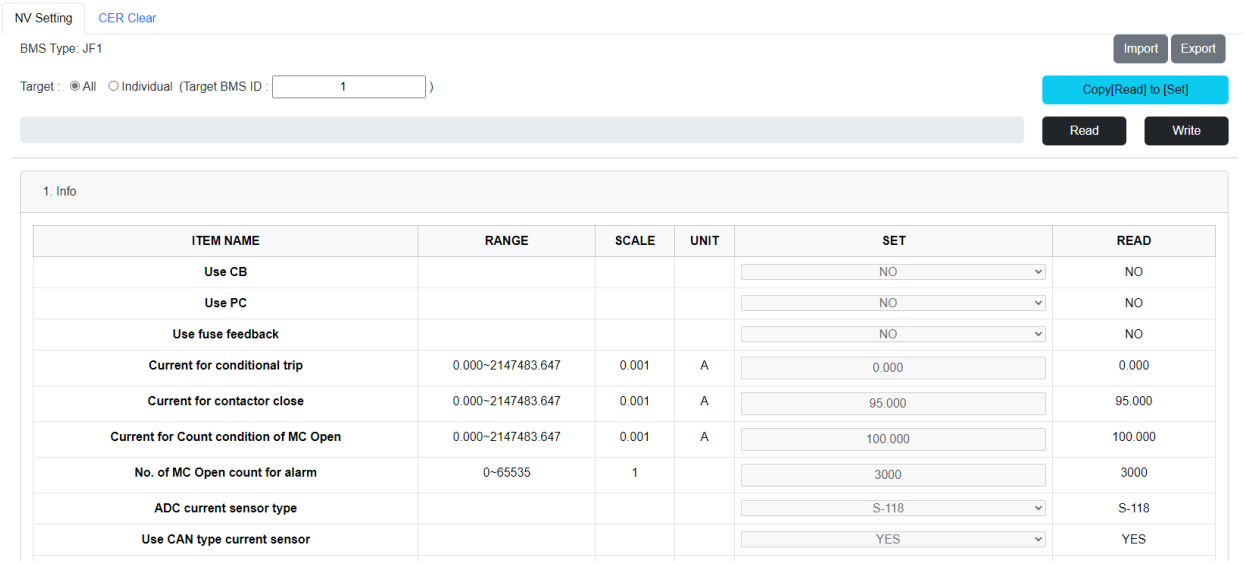


Figure 28. NV Setting Icon

1) NV Setting

NV configuration is a function that sets up a lower RBMS. When performing SAT, the system can operate normally only when it is set up with the BSC Configurator.



ITEM NAME	RANGE	SCALE	UNIT	SET	READ
Use CB				NO	NO
Use PC				NO	NO
Use fuse feedback				NO	NO
Current for conditional trip	0.000~2147483.647	0.001	A	0.000	0.000
Current for contactor close	0.000~2147483.647	0.001	A	95.000	95.000
Current for Count condition of MC Open	0.000~2147483.647	0.001	A	100.000	100.000
No. of MC Open count for alarm	0~65535	1		3000	3000
ADC current sensor type				S-118	S-118
Use CAN type current sensor				YES	YES

Figure 29.NV Setting View

(1) Target: Select the target to execute NV Read/Write. If you select Target All, the same operation is performed on all connected RBMS, and if you select Individual, you can select the Rack to execute the command and proceed individually.

(2) NV Group: Displays the NV Category group to execute the NV Write command. You can select the group you want to execute the command. (Default: All)

(3) Item List View : Shows the NV item list. Set column means the value input when writing, and in

case of read column, it displays the NV value received from BMS. Changing the values of items is not allowed.

(4) NV Command: You can write NV by setting the value in the [SET] column in the Item List View and pressing the [Write] button. If you press the [Read] button, you can read the NV value received from the BMS. The read value can be checked in [READ] Column. To check consistency, if the value of the [SET] column and the value of the [READ] column are different, the value of the [READ] column is highlighted in red.

(5) Progress Bar: You can check the progress of the NV command in a bar-shaped graph.

(6) Copy [Read] to [Set] : Copy the [Read] column values to the [Set] column.

(7) Import/Export: You can import or export NV setting values as a file. When importing/exporting, update or save the value in [Set] Column.

2) CER Clear

CER stands for Critical Event Record and is recorded when a fatal error situation occurs leading to a state in which the battery cannot be operated. When this situation (MDVF, RDVF, Emergency Mode, etc.) occurs, even if restart of the BMS is performed, the battery cannot be operated. In the event of an actual critical event, the LG Energy Solution engineer can use this function to check the rack where the event occurred, and can release the critical event after checking the pack and taking proper action.

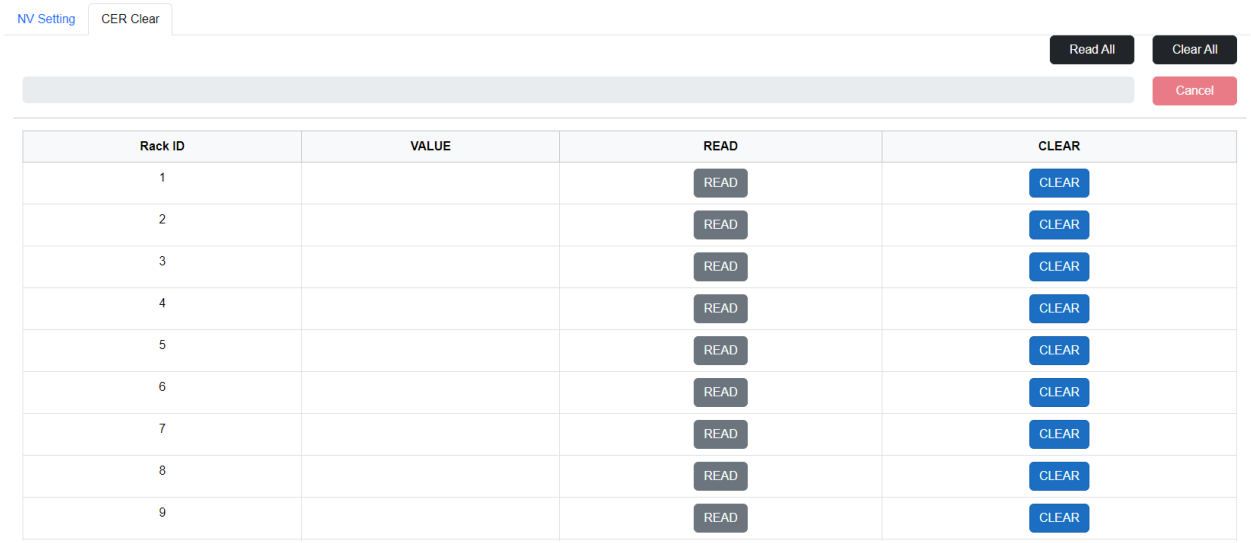


Figure 30. CER Clear View

(1) CER Value List View: Check the CER value of each rack. CER Value can be read or cleared for each rack through the Read and Clear button.

(2) Read All/Clear All : The Read/Clear command is executed across the board.

(3) Progress Bar : The progress of the CER command can be checked with a bar-type graph.

4. Log Files

4.1. Data Log

Log files, as described below, are created and stored every day.

The directory for System Log files is /mnt/d/log/Gen3BSC/Section1/Base/YYYYMMDD

(where YYYYMMDD is the date in the format YYYYMMDD).

The directory for Pack Log files (1-second interval) is /mnt/d/log/Gen3BSC/Section1/Pack/YYYYMMDD.

- Flex_BSC_Section_S1_YYYYMMDD.csv: The battery section's operating information
- Flex_BSC_Rack_S1R##_YYYYMMDD.csv: A battery rack's operating information (##:Rack No.)
- Flex_BSC_Packs_S1R##_YYYYMMDD.csv: A battery pack's operating information of the Rack(##: Rack No.)
- Flex_BSC_SystemEvent_S1_YYYYMMDD.txt: Event History

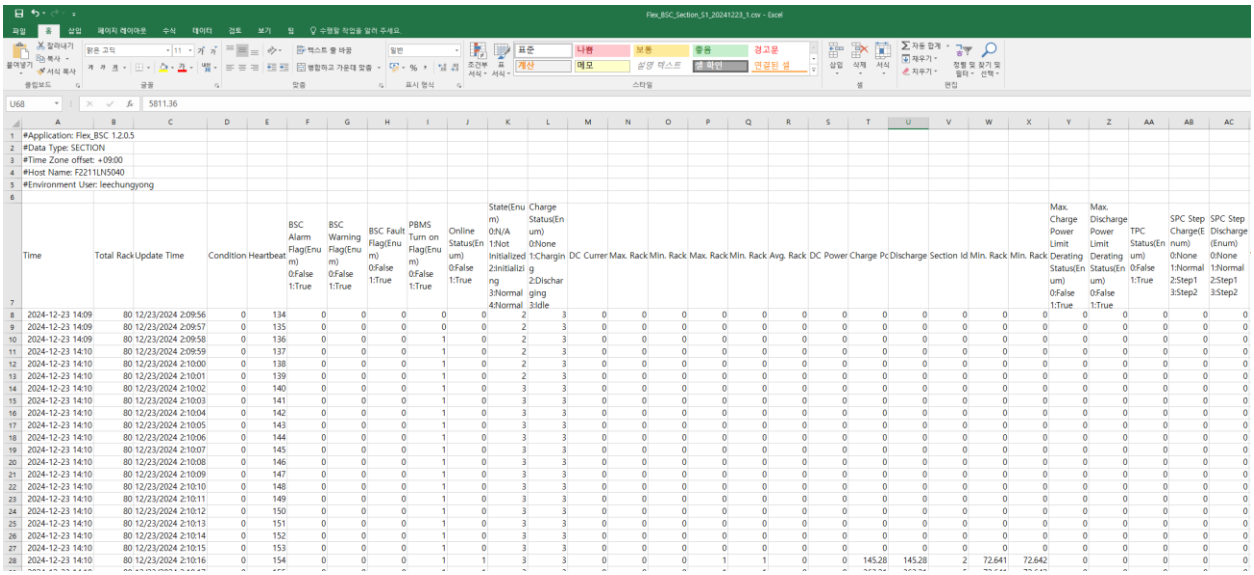


Figure 31. Log File Example

4.2. **META Log**

When BMS is connected, META data received from BMS is saved in a file.

The directory of Meta log files is /mnt/d/log/Gen3BSC/Section1/Base/

File Name : Flex_BSC_Nameplate_S1_YYYYMMDD.txt



Figure 32 Meta Log File Example